CLOUD DATA ENGINEER

AWS S3 Static Hosting

Project : Amazon S3, EC2, VPC, Linux and Nginx

Item	Details
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Program	Cloud Data Engineer
Project	Static website hosting on AWS (S3 and EC2 + Nginx)

1. Project Overview

This project documents how a static HTML/CSS portfolio website was hosted on Amazon Web Services. A web server was launched on Amazon EC2 (Amazon Linux 2023), connected to through EC2 Instance Connect, configured with Nginx, and tested in a browser. Amazon S3 bucket creation and object uploads are also documented, along with the Linux commands practised during the project.

Scope covered in this document:

- Amazon S3 bucket creation and uploaded objects
- Amazon EC2 instance launch, AMI, instance type and key pair
- VPC creation settings
- Security Group inbound and outbound rules
- Linux server connection (EC2 Instance Connect and SSH client) and commands
- Nginx service status and website deployment
- Website testing and final output
- Comparison of S3 hosting versus EC2 + Nginx hosting

2. Architecture

The diagram shows the hosting path demonstrated by the screenshots: a browser reaches Nginx on the EC2 instance through the Security Group, which allows HTTP (port 80) and SSH (port 22).

```
Browser (HTTP, port 80)
       |
   Internet
       |
+---------------- AWS Region ----------------------+
|  VPC                                             |
|   +-- Subnet                                     |
|        +-- EC2 instance (Amazon Linux 2023)      |
|             - Nginx web server                   |
|             - Portfolio: index.html, style.css,  |
|               about.html                         |
|             - Security Group: 22 and 80 inbound  |
+--------------------------------------------------+

Amazon S3: bucket creation and uploads are documented;
static website hosting.
```

3. Amazon S3

3.1 Bucket creation

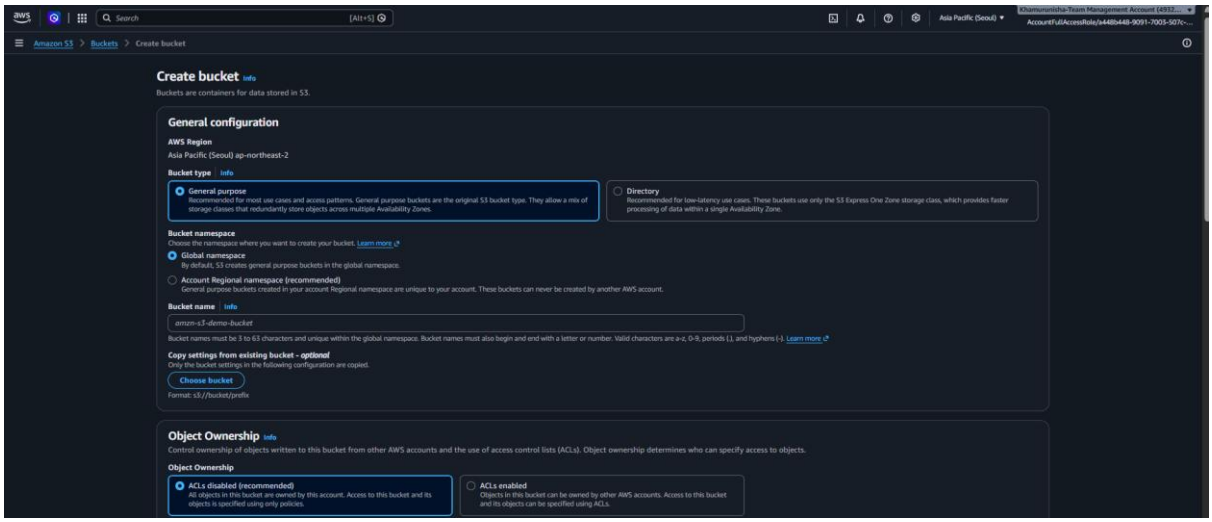


Figure 1: *Amazon S3 Create bucket: general configuration and object ownership.*

3.2 Uploaded objects

A bucket named college-admission-docs is shown with an incoming/ prefix containing three Standard-class objects: APP-1003.json, Cloud Admission System.txt and sample-application.json.

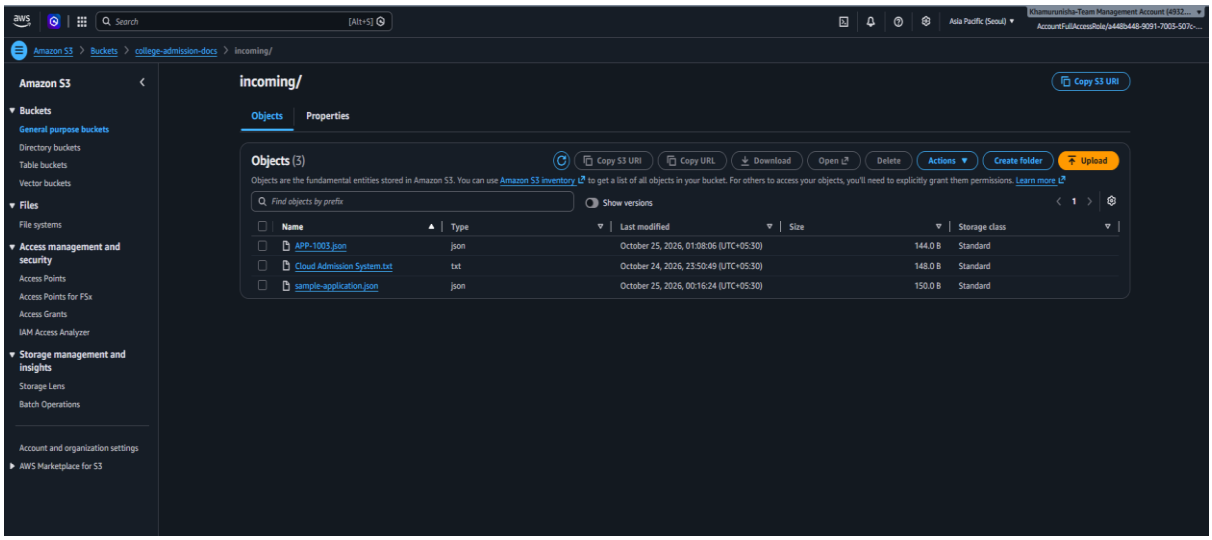


Figure 2: *Objects stored under the incoming/ prefix of an S3 bucket.*

4. Amazon EC2

4.1 Instance launch and AMI selection

The instance was named MY WEB SERVER. Under Application and OS Images, the Quick Start tab is open with the Amazon Linux 2023 AMI selected, which is marked Free tier eligible.

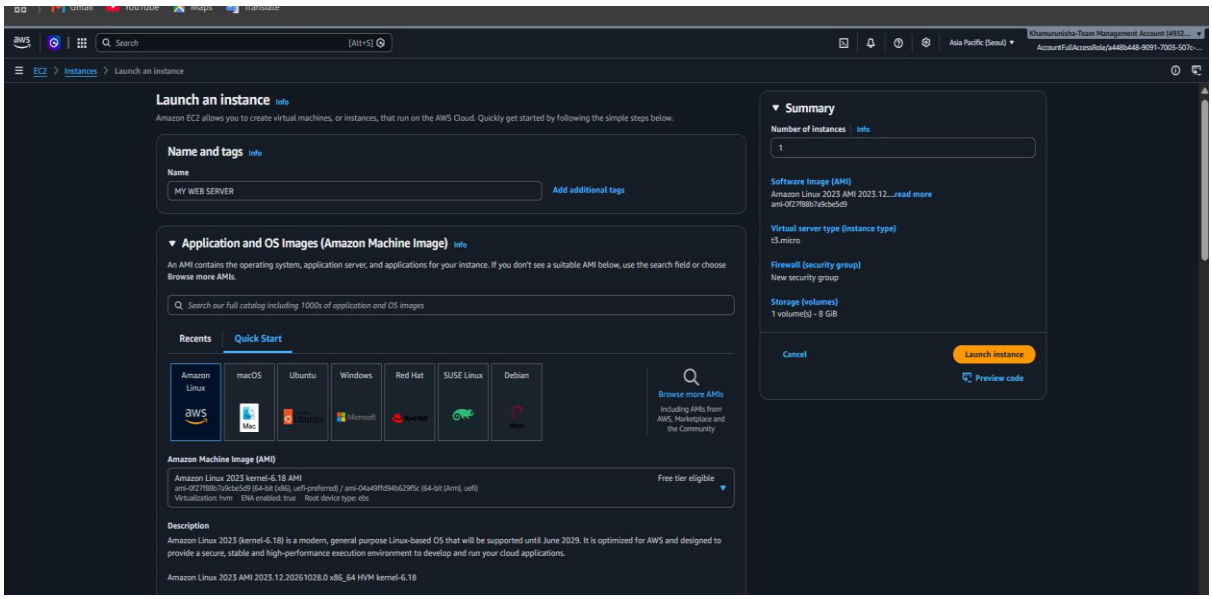


Figure 3: *EC2 Launch an instance: name and Amazon Machine Image.*

4.2 Instance type, key pair and network settings

The instance type is t3.micro (Free tier eligible). A key pair named MYKEYPAIR is created with the RSA type and the .pem private key format, which is used with OpenSSH.

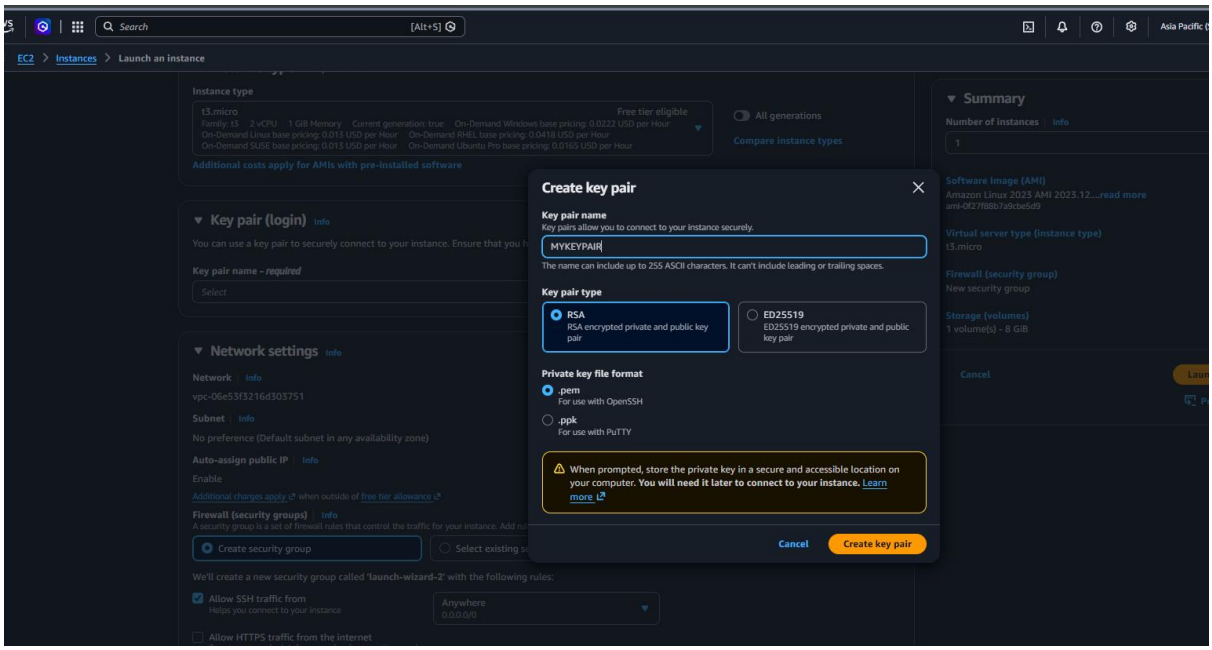


Figure 4: *Instance type, Create key pair dialog and network settings.*

4.3 VPC settings

The Create VPC page shows: Resources to create set to "VPC only", Name tag my-vpc-01, IPv4 CIDR set by manual input as 10.0.0.0/24, No IPv6 CIDR block, and Default tenancy.

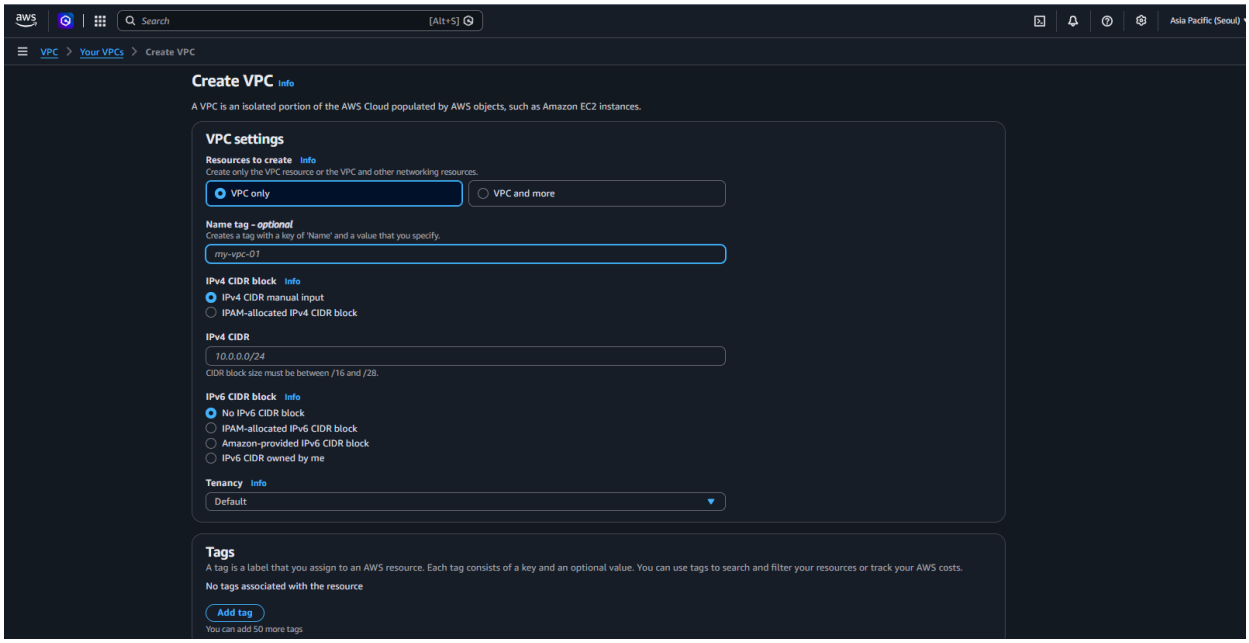


Figure 5: *Create VPC with the my-vpc-01 name tag and 10.0.0.0/24 CIDR.*

4.4 Connecting to the instance

The Connect to instance page shows MY WEB SERVER in the Running state with 3/3 status checks passed. EC2 Instance Connect is selected, a public IPv4 address is chosen, and the username is ec2-user.

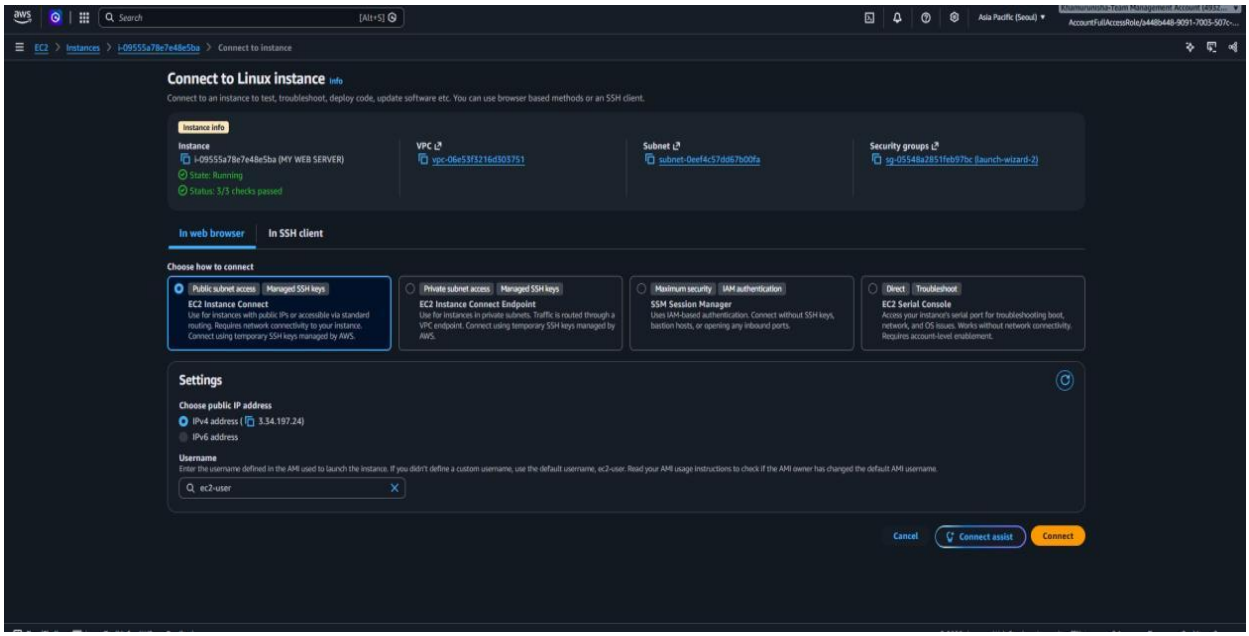


Figure 6: *Connect to Linux instance using EC2 Instance Connect (web browser).*

The SSH client tab lists the key pair (MYKEYPAIR), the instance public DNS and username, and the standard connection steps: locate the private key, restrict its permissions with chmod 400, and connect with ssh -i using the public DNS name.

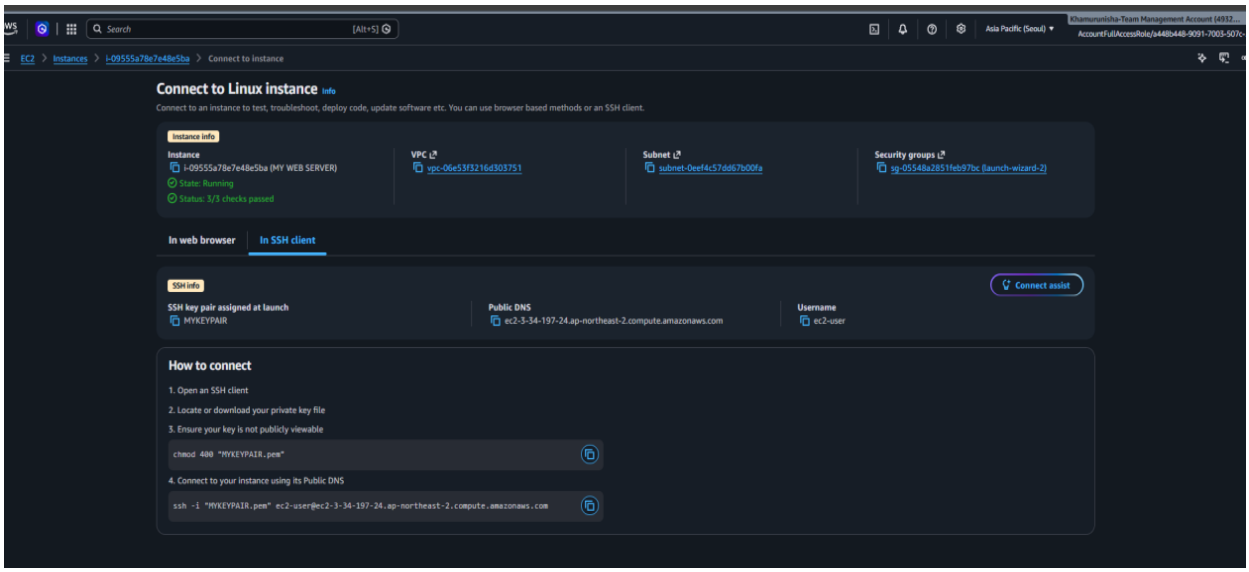


Figure 7: *Connect to Linux instance using an SSH client.*

4.5 Security Group rules

The instance Security details show the following rules for the launch-wizard security group.

Direction	Port range	Protocol	Source / Destination
Inbound (SSH)	22	TCP	0.0.0.0/0
Inbound (HTTP)	80	TCP	0.0.0.0/0
Outbound	All	All	0.0.0.0/0

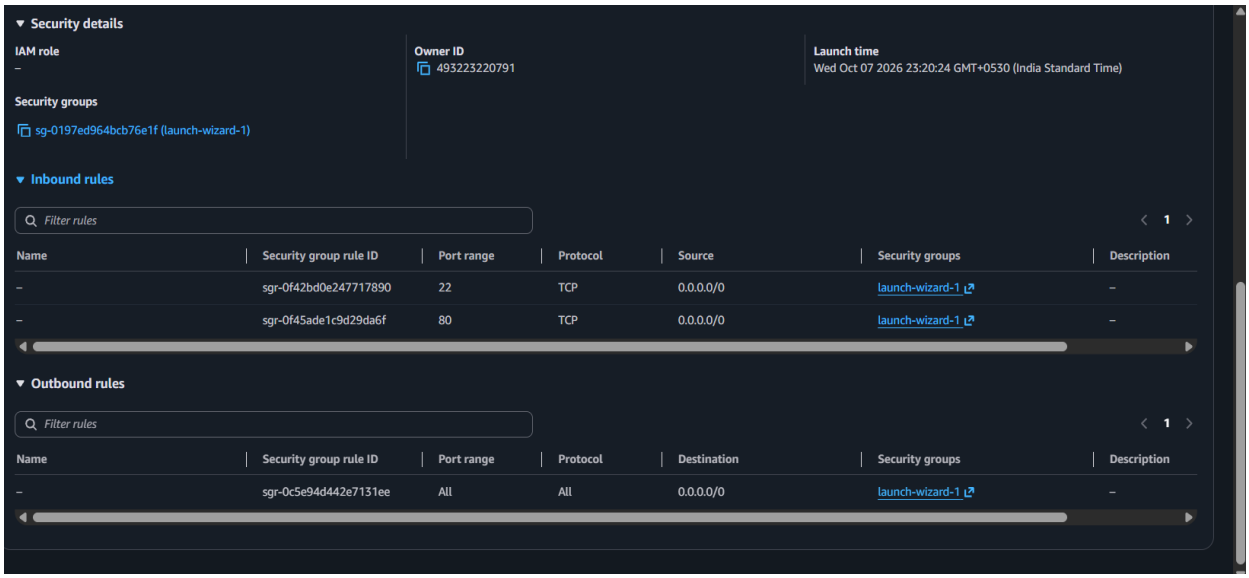


Figure 8: *Security details: inbound and outbound rules.*

5. Linux

5.1 EC2 server session

The browser terminal connected through EC2 Instance Connect shows the Amazon Linux 2023 welcome banner and the commands who am i, nginx -v and sudo systemctl status nginx --no-pager, all run as ec2-user.

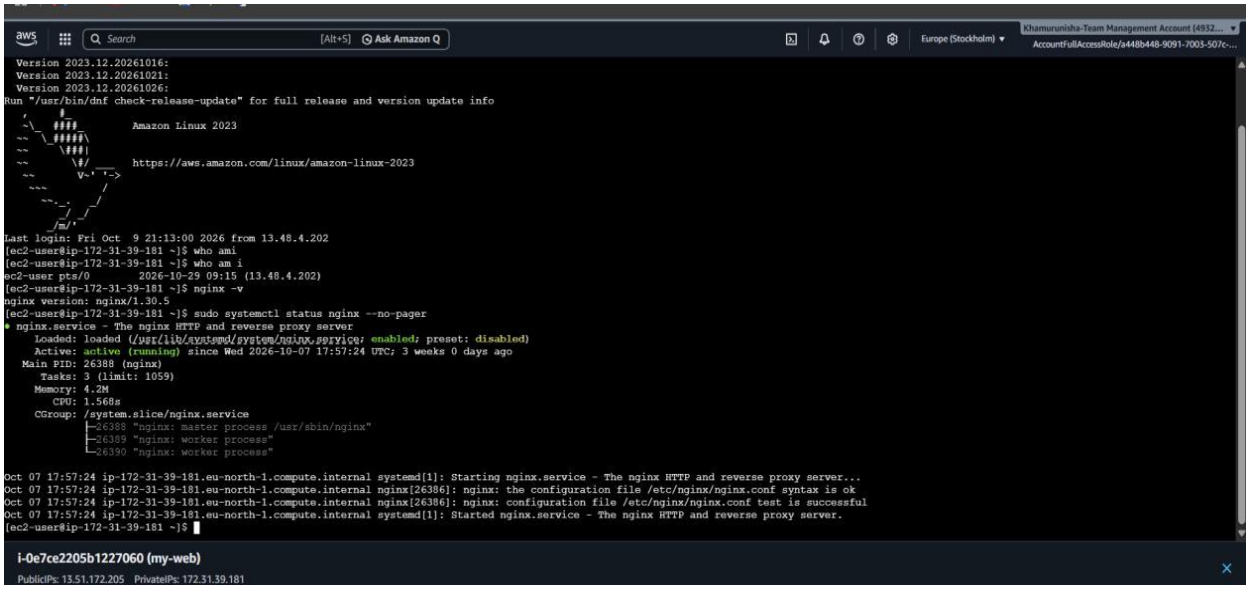


Figure 9: *EC2 Instance Connect terminal on Amazon Linux 2023 with Nginx status.*

5.2 Local Ubuntu practice (WSL)

Basic Linux commands were also practised in a local Ubuntu terminal running on Windows Subsystem for Linux (WSL2). This session covers pwd, whoami, date, uname -a, ls -lah, creating a directory and a file, writing to it with echo, reading it with cat, and checking disk usage with df -h.

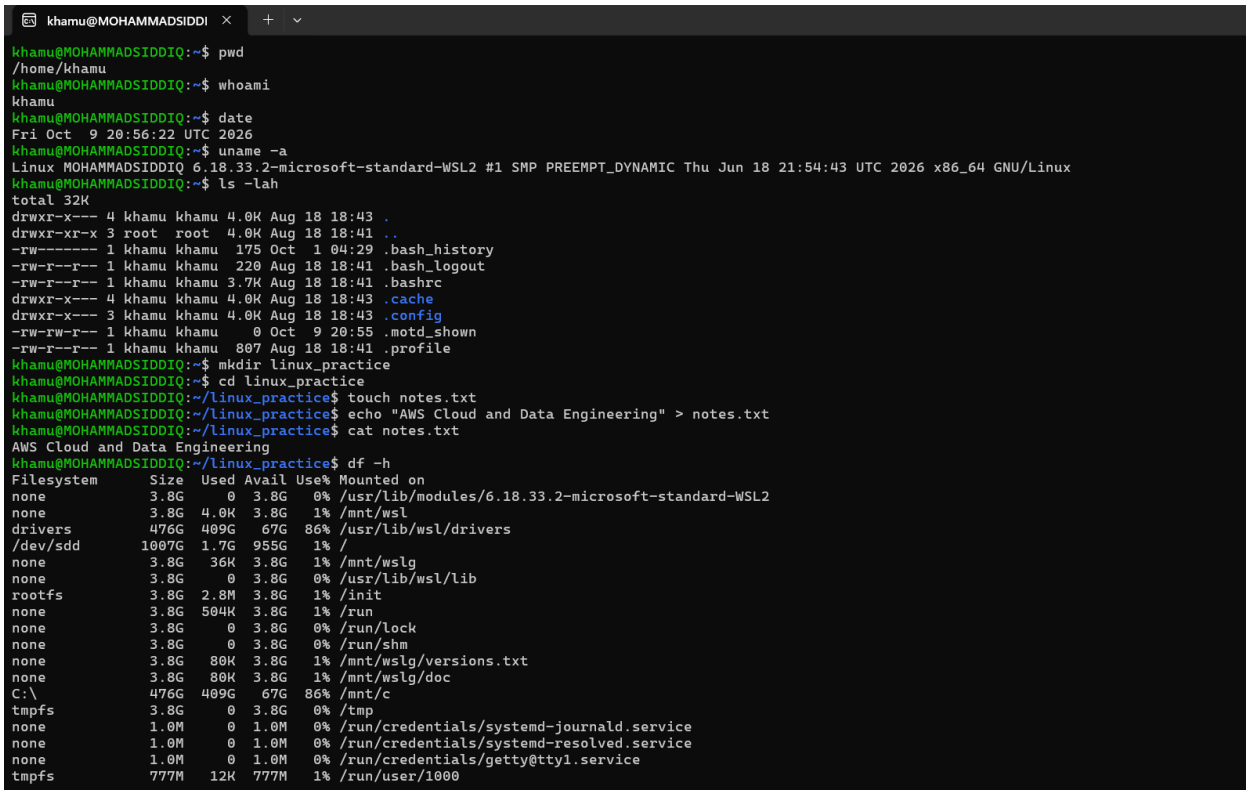


Figure 10: *Local Ubuntu (WSL2) terminal: basic commands and disk usage.*

The next session creates the portfolio project folder with mkdir -p, creates index.html, style.css and about.html with touch, opens each in nano, and starts a local test server with python3 -m http.server 8000. The server log shows the pages being requested and returned successfully (status 200 and 304).

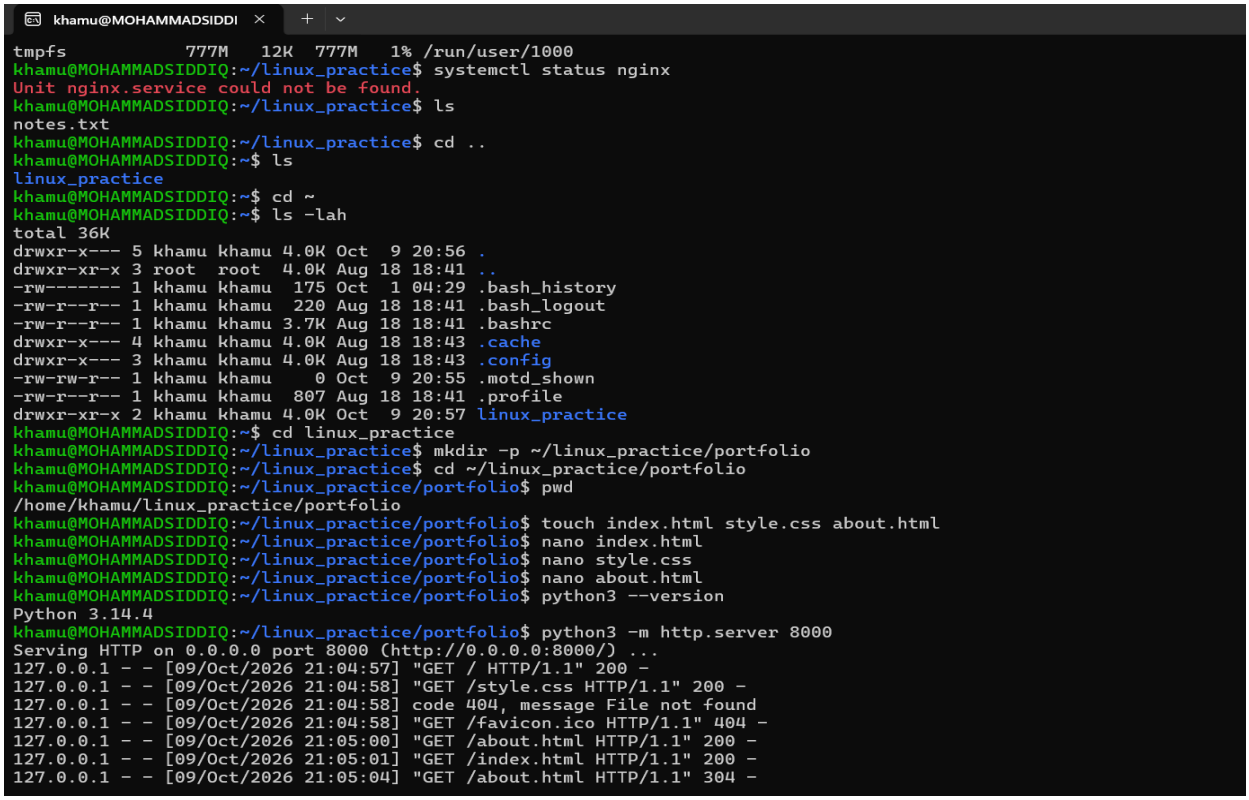


Figure 11: *Local Ubuntu (WSL2): portfolio folder, file creation and Python test server.*

5.3 Command reference

Basic Linux commands

No.	Command	Use
1	`pwd`	Displays the current working directory.
2	`whoami`	Displays the current logged-in username.
3	`date`	Displays the system date and time.
4	`uname -a`	Displays Linux kernel and system information.
5	`ls -lah`	Lists files, including hidden files, with details.
6	`ls`	Lists files and directories in the current location.
7	`cd linux_practice`	Changes the current directory to linux_practice.
8	`cd ..`	Moves to the parent directory.
9	`cd ~`	Navigates to the user's home directory.

Directory and file management

No.	Command	Use
10	`mkdir linux_practice`	Creates a directory named linux_practice.
11	`mkdir -p ~/linux_practice/portfolio`	Creates the portfolio directory and any missing parent directories.
12	`touch notes.txt`	Creates an empty file named notes.txt.
13	`touch index.html style.css about.html`	Creates three website files.
14	`nano index.html`	Opens the HTML file for editing in Nano.
15	`nano style.css`	Opens the CSS file for editing in Nano.
16	`nano about.html`	Opens the About page for editing in Nano.
17	`echo "AWS Cloud and Data Engineering" > notes.txt`	Writes text to notes.txt, replacing existing content.
18	`cat notes.txt`	Displays the contents of notes.txt.

Website project steps

No.	Command or action	Use
19	`mkdir -p ~/linux_practice/portfolio`	Creates the website project directory.
20	`cd ~/linux_practice/portfolio`	Opens the website project directory.
21	`touch index.html style.css about.html`	Creates the homepage, stylesheet and About page.
22	`nano index.html`	Creates or edits the website homepage HTML.
23	`nano style.css`	Creates or edits the website styling.
24	`nano about.html`	Creates or edits the About page HTML.
25	`ls -lah`	Verifies files and directories, including hidden files.
36	`cat notes.txt`	Verifies the text saved in the notes file.

6. Nginx

Reference commands. The following are the standard commands for Nginx on Amazon Linux 2023. The install step is not captured in the screenshots, so these are given for reference only.

```
sudo dnf install nginx -y            # install
sudo systemctl enable --now nginx    # start now and at boot
nginx -v                             # show version
sudo nginx -t                        # test configuration
sudo systemctl status nginx --no-pager
sudo systemctl reload nginx          # apply changes
```

The default Nginx web root on Amazon Linux 2023 is /usr/share/nginx/html.

7. Website Deployment and Testing

The portfolio website was opened in Google Chrome using the EC2 public IP address over HTTP (the browser marks it "Not secure" because no HTTPS certificate is configured). The home page shows the name Mohammad Siddiq Shaik, the title Computer Science Engineering Student, a short tagline, View Projects and Contact Me buttons, and a navigation bar with Home, About, Education, Skills, Experience, Projects and Contact.

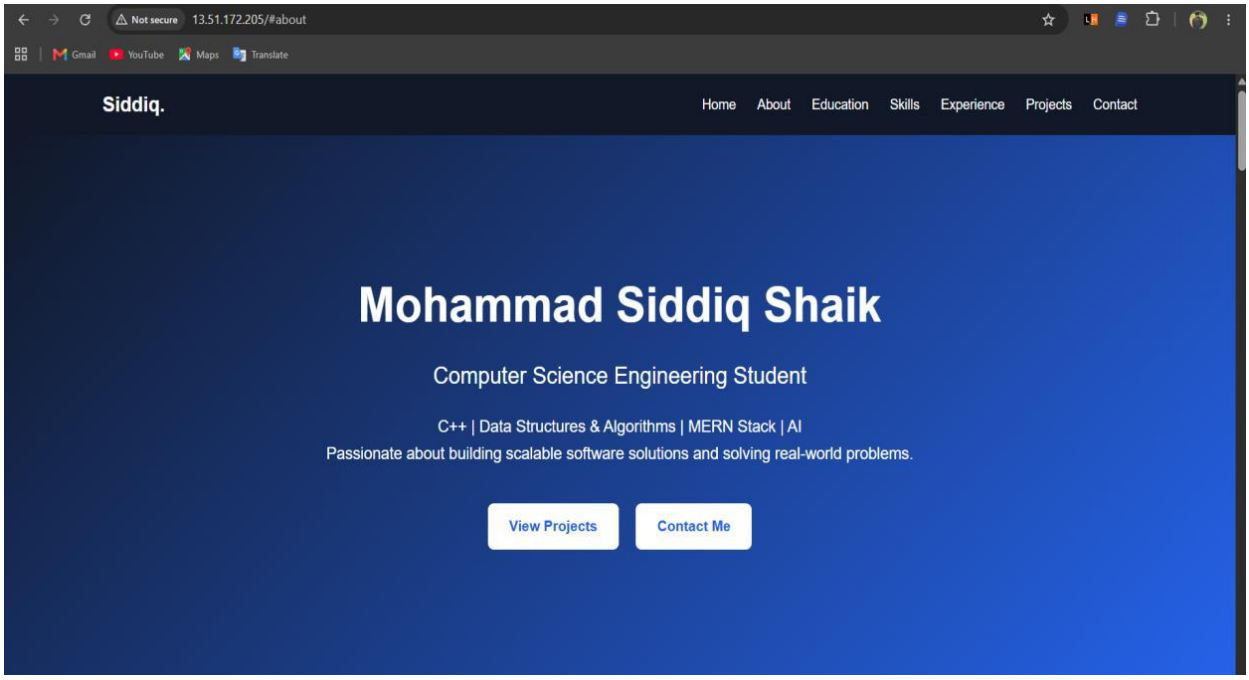


Figure 12: *Final output: portfolio home page served by Nginx.*

The Technical Skills section lists the technologies used, including C++, Python, JavaScript, SQL, HTML5, CSS3, React.js, Node.js, Express.js, REST APIs, JWT, OAuth 2.0, MongoDB, Mongoose, SQLite, Redis, Google Gemini API, NLP, YOLOv8, Git, GitHub, VS Code, Postman, DSA, OOP, DBMS, Operating Systems, Computer Networks and SDLC.

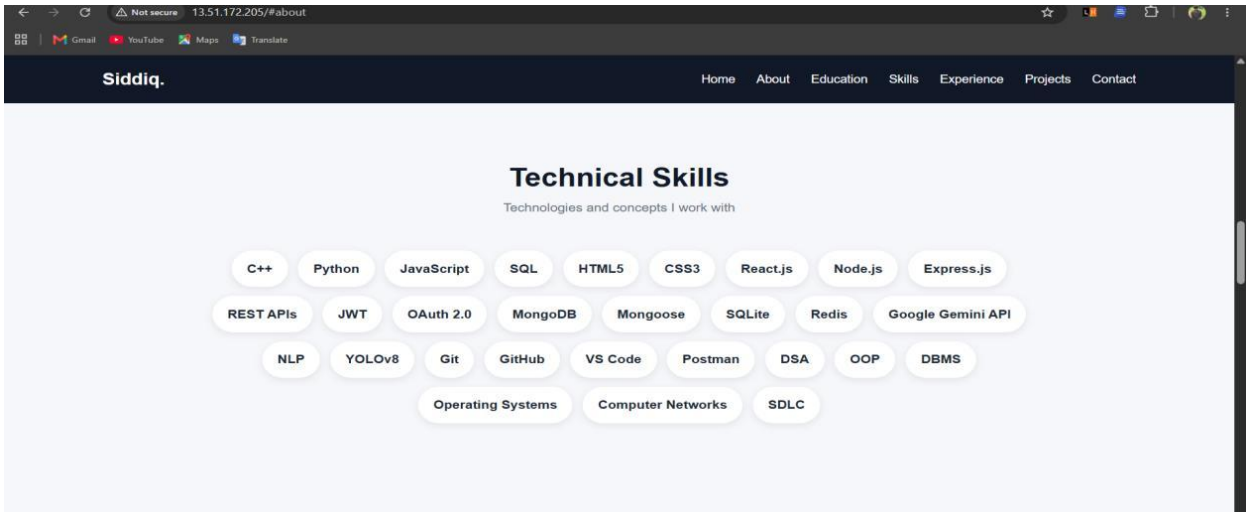


Figure 13: *Final output: Technical Skills section.*

The Achievements and Contact sections confirm that the later part of the page loads correctly, including the navigation anchors (the URL ends with #contact).

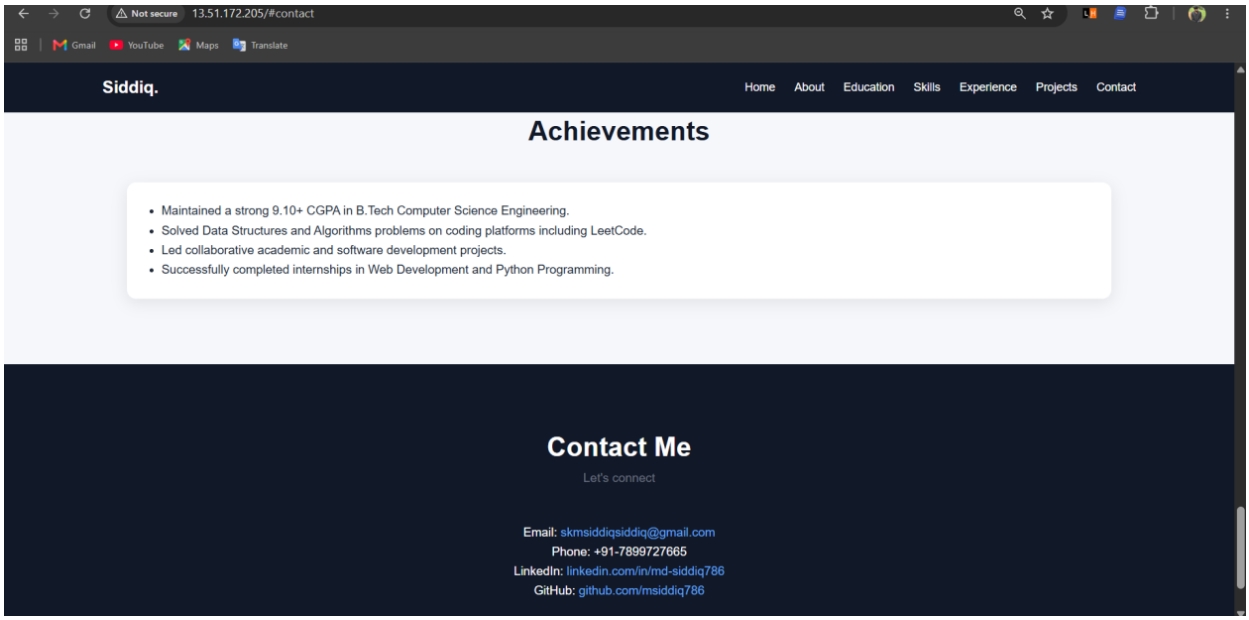


Figure 14: *Final output: Achievements and Contact sections.*

Test result: the pages load successfully from the EC2 instance through Nginx, and in-page navigation works.

Conclusion

This project demonstrates launching an Amazon Linux 2023 EC2 instance, configuring a key pair and Security Group, connecting through EC2 Instance Connect, confirming the Nginx service is active and running, and serving a static portfolio website to a browser. It also documents S3 bucket creation and object listing, the VPC creation settings, and a Linux command practice log.